

1 input

a **line** in 2D

$$\hat{y} = \beta_0 + \beta_1 x_1$$



2 inputs

a **plane** in 3D

$$\hat{y} = \beta_0 + \beta_1 x_1 + \beta_2 x_2$$



n inputs

a **hyperplane** in $(n+1)$ D

$$\hat{y} = \beta_0 + \beta_1 x_1 + \cdots + \beta_n x_n$$

β_0 : intercept $\beta_1, \dots, \beta_n$: one coefficient (weight) per input column